

Indian Institute of Technology Bombay

PH 544 - Quiz 1

General relativity

Date: 16-02-2026

Marks 15 = Weightage 10 points

LC-302

Time: 17:40-18:40

1. If you drive towards a red traffic light with sufficiently high speed, you would observe the traffic light shifted upward in frequency. With sufficient speed, the red light might turn into a green light.

(a) With what speed must you approach a red traffic light ($\lambda = 625 \text{ nm}$) so that it appears to be a green light ($\lambda = 500 \text{ nm}$)? [2]

(b) Assuming the speed from Part (a), what wavelength would you observe for a yellow traffic light that has a rest wavelength of 570 nm ? [2]

$$v = v_0 \sqrt{\frac{c+v}{c-v}}$$

2. A paraboloid is described by the relations:

$$x = 2at \cos \phi ; \quad y = 2at \sin \phi ; \quad z = at^2$$

where a is a constant and t is a parameter.

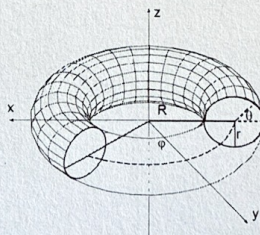
(a) Show that the line-element of a paraboloid is [2]

$$ds^2 = 4a^2(1+t^2)dt^2 + 4a^2t^2d\phi^2$$

(b) Obtain Geodesics of the paraboloid. [2]

3. Show that $g_{,n}^{im} = -g^{mk}g^{il}g_{kl,n}$ where $g_{,n}^{im} = \frac{\partial g^{im}}{\partial x^n}$. [2]

4. Consider a donut (torus). The 2D metric that describes the surface of the donut is given by: $g_{\theta\theta} = (R + r \cos \phi)^2$, $g_{\phi\phi} = r^2$, and $g_{\theta\phi} = g_{\phi\theta} = 0$. r is the radius of the cross-section of the torus.



(a) Compute the Christoffel symbols. [3]

$$\Gamma_{ij}^k = \frac{1}{2}g^{lk} \left(\frac{\partial g_{il}}{\partial q^j} + \frac{\partial g_{jl}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^l} \right)$$

(b) Compute the Riemann tensor for the Torus. [2]